

Muhammad Saram Hassan

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EDUCATION

Lahore University of Management Sciences (LUMS)

Aug 2022 - May 2026 (Expected)

Bachelor of Science in Computer Science

- **Dean's Honor List (2022-2025)**
- **Relevant Graduate Coursework:** Advanced Topics in ML[†], Foundations of Generative AI[†], Topics in Network Security[†], Distributed Systems[†]

SELECTED PUBLICATIONS

Data Doping or True Intelligence? Evaluating the Transferability of Injected Knowledge in LLMs

[arXiv:2505.17140](https://arxiv.org/abs/2505.17140) | *EMNLP Findings 2025* | Acceptance Rate: **17.35%**

Essa Jan, Moiz Ali, **Muhammad Saram Hassan**, Fareed Zaffar, Yasir Zaki.

RESEARCH EXPERIENCE

Research Assistant - TPI Lab, LUMS & CommNets Lab, NYUAD

Jan 2025 – Jul 2025

Supervised by: [Dr. Fareed Zaffar \(LUMS\)](#) [Dr. Yasir Zaki \(New York University Abu Dhabi\)](#)

Knowledge Retention and Transferability in Fine-Tuned LLMs [EMNLP Findings 2025]

- **Research Question:** How does the task type (e.g., token-to-token mapping vs. content understanding) and model size influence the knowledge retention, integration, and generalizability during fine-tuning?
- Designed controlled experiments isolating task format as predictor of knowledge integration robustness during supervised fine-tuning
- Evaluated 5 model families (across 1.5B-72B parameters) on surface vs. comprehension tasks, comparing retention and scaling trends.
- Discovered that comprehension-based tasks (e.g question answering) achieved 48% retention versus 17% for surface-level mapping tasks (e.g translation), revealing that deeper semantic integration enhances transferability.
- Observed consistent scaling improvements in knowledge retention from 38% to 72% across models for all tasks, revealing systematic advantages of scale for conceptual knowledge integration.
- Published at **EMNLP Findings 2025**

Research Intern - Stanford Research Institute (SRI) International

Aug 2025 – Present

Supervised by: [Dr. Ashish Gehani](#) and [Dr. Ramneet Kaur](#)

Quantifying Conceptual Understanding in LLMs

- **Research Question:** Do language models demonstrate true conceptual understanding that is robust across tasks and contexts, or do they rely on spurious correlations that break down under distribution shifts?
- Developing model-agnostic evaluation framework to distinguish genuine conceptual understanding from brittle pattern matching that fails under distribution shift
- Designed dual-metric system: Behavioral Consistency Score (output stability across paraphrased inputs) and Representation Consistency Score (embedding geometry coherence)
- Introduced Concept Coherence Index (CCI) unifying behavioral and mechanistic evaluation; extended with Latent Direction Score (LDS) for causal probing of concept representations via Sparse Autoencoders
- Targeting **ICML 2026**

Research Assistant - Security & Privacy Lab, LUMS

Supervised by: [Dr. Fareed Zaffar \(LUMS\)](#) and [Dr. Taha Khan \(CMU CyLab\)](#)

Adversarial Robustness of LLM-Based Web Agents

- **Research Question:** How vulnerable are autonomous LLM-based web agents to adversarial manipulation by malicious websites, and what behavioral signatures enable detection of agent traffic?
- Leading Systematization of Knowledge (SoK) on vulnerabilities in autonomous web agents, developing taxonomy distinguishing agentic risks from traditional crawler exploits.
- Built adversarial testbed demonstrating practical attacks: MCP poisoning enabling persistent adversarial directives across sessions, indirect prompt injection via DOM manipulation, visual spoofing.

- Developed behavioral fingerprinting framework analyzing HTTP headers, DOM traversal patterns, timing signatures, and robots.txt/CAPTCHA compliance to detect LLM agent traffic
- Discovered that LLM browsers bypass robots.txt while masquerading as humans, triggering ad-auctions and generating revenue without detection—surfacing regulatory gaps in agent deployment

Research Assistant - CITY Lab @ LUMS

Sep 2024 – May 2025

Supervised by: *Dr. Muhammad Tahir (Associate Professor, LUMS)*

Bias-Aware Interpretable Pruning for Robust Vision Models

- **Research Question:** How does neural network compression affect inductive biases (texture, shape, color), and can we selectively preserve or remove biases to improve OOD robustness?
- Developed bias-aware pruning framework combining property-amplified augmentation with gradient-saliency and magnitude-based importance estimation for CNNs and ViTs.
- Implemented interpretable pruning methodology for selective bias retention/removal; introduced filter-locking mechanism preserving domain-invariant features.

SELECTED PROJECTS

LLM Password Generation: Security, Usability, and Policy Compliance

Present

- Evaluating LLM-generated passwords across security (entropy, resistance to guessing attacks, similarity to breached passwords), usability (memorability), and policy compliance dimensions
- Analyzing how model size, temperature, and prompting strategies affect generation quality and consistency
- Investigating failure modes: policy violation patterns, hallucination rates, and exploitable regularities in generated passwords
- Collaboration with **CMU CyLab (Dr. Taha Khan)**

Voice AI for Regional Languages | Uplift AI Partnership (YC startup)

Jun 2024 - Jul 2024

- Led a cross-functional team of **100+ students** with **Uplift AI** to collect and curate **35,000+** regional language speech samples, directly contributing to the startup securing funding from Y-Combinator.
- Designed quality control and preprocessing pipelines ensuring **95%+** transcription accuracy, directly contributing to UpliftAI's successful seed funding round from Y-Combinator.

Model Compression via Structured and Unstructured Pruning

Aug 2024

- Applied L2-norm unstructured and channel-wise structured pruning to VGG16, achieving **65.9%** post-fine-tuning accuracy (**7.5% lift**) from the baseline.
- Implemented PTQ and QAT across multiple bit-widths (FP16, BF16, INT8, INT4), visualizing accuracy–precision scaling trends.

Knowledge Distillation Techniques via Feature Alignment and KL Divergence

- Compared Hint-based Distillation, Logit Matching, Label Smoothing, and Decoupled KD methods under domain shifts and adversarial attacks using GradCAM visualizations.
- Hint-based Distillation achieved the highest student accuracy at 89%, while Logit Matching minimized the KL-divergence the most, indicating stronger teacher–student alignment.

Raft-Based Fault-Tolerant Distributed Key-Value Store

Oct 2024 - Dec 2024

- Implemented a complete Raft consensus module in Go with leader election, log replication, commit indexing, snapshotting, and log compaction, ensuring consistent state across replicas and fast recovery after failures.
- Designed and built a distributed key value store providing strong consistency, automatic crash recovery, and durable state persistence across replicas.

TEACHING EXPERIENCE

Supported 300+ students across theory and programming courses, leading tutorials and labs that strengthened core reasoning and coding skills. Designed and delivered two applied ML and astronomy courses taken by over 100 students.

CS-210: Discrete Mathematics *Dr. Malik Jahan, LUMS*

CS-100: Computational Problem Solving *Dr. Fareed Zaffar, LUMS*

Fundamentals of Machine Learning *LUMS Science School & UpliftAI*

Exploring the Universe with Code: CS for Astronomy *FAST University*

TECHNICAL SKILLS

Languages: Python, Haskell, Go, C/C++, JavaScript, C#

Frameworks: React, Node.js, WordPress, PyTorch, Hugging Face (Transformers, Diffusers, timm, unsloth)

Tools: Git, Docker, VS Code, LLVM, LangChain,